

# “Calculus 3”

## Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 11

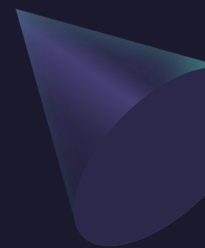


# Any Reminders? Any Questions?

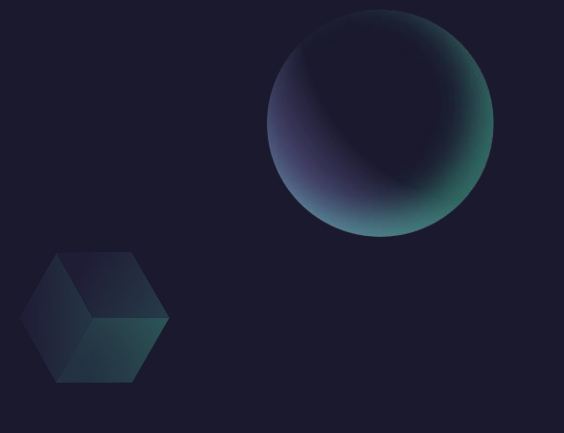
- I have regular office hours Mondays via Webex 1-2pm
- I have regular office hours Thursdays in-person 3:30-4:30pm
- Calc 3 Calc Night: MONT 104 at 6:30-8:30pm on Thursdays!



*ALVARO:* Start the recording!



# “Calculus 3”



## Multi-Variable Calculus

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**MORE ON**



## Double Integrals over Regions





# Today – Double Integrals in Regions!

- 
- General Regions
  - Regions of Type I and II
  - Changing the Order of Integration
  - Properties of Double Integrals

# Regions of Type I and II

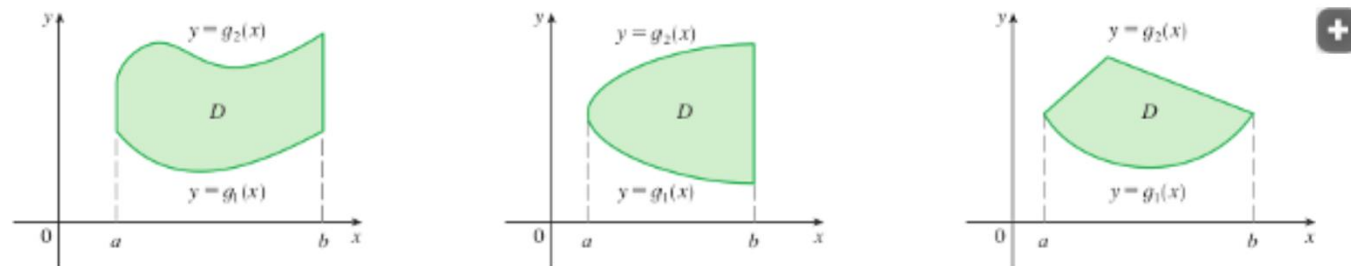
A plane region  $D$  is said to be of **type I** if it lies between the graphs of two continuous functions of  $x$ , that is,

$$D = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

where  $g_1$  and  $g_2$  are continuous on  $[a, b]$ . Some examples of type I regions are shown in [Figure 5](#).

**Figure 5**

Some type I regions



# Regions of Type I and II

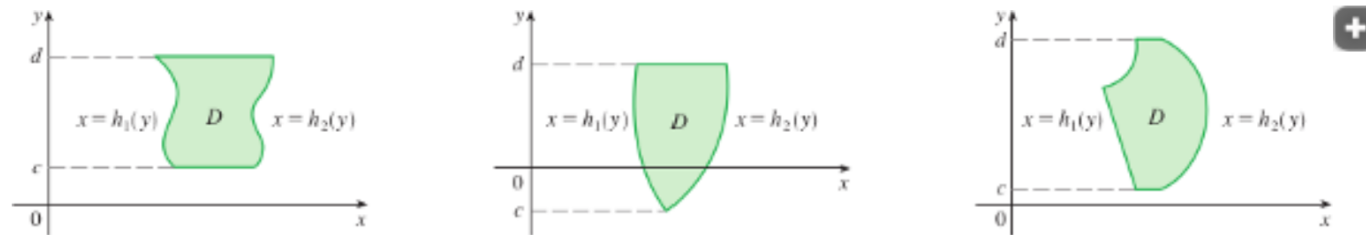
We also consider plane regions of **type II**, which can be expressed as

$$D = \{(x, y) \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$$

where  $h_1$  and  $h_2$  are continuous. Three such regions are illustrated in [Figure 7](#).

**Figure 7**

Some type II regions



# Integrals over Regions of Type I

**3** If  $f$  is continuous on a type I region  $D$  described by

$$D = \{(x, y) \mid a \leq x \leq b, \ g_1(x) \leq y \leq g_2(x)\}$$

then

$$\iint_D f(x, y) \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx$$



# Integrals over Regions of Type II

**4** If  $f$  is continuous on a type II region  $D$  described by

$$D = \{(x, y) \mid c \leq y \leq d, \ h_1(y) \leq x \leq h_2(y)\}$$

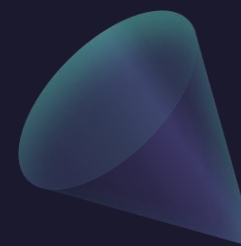
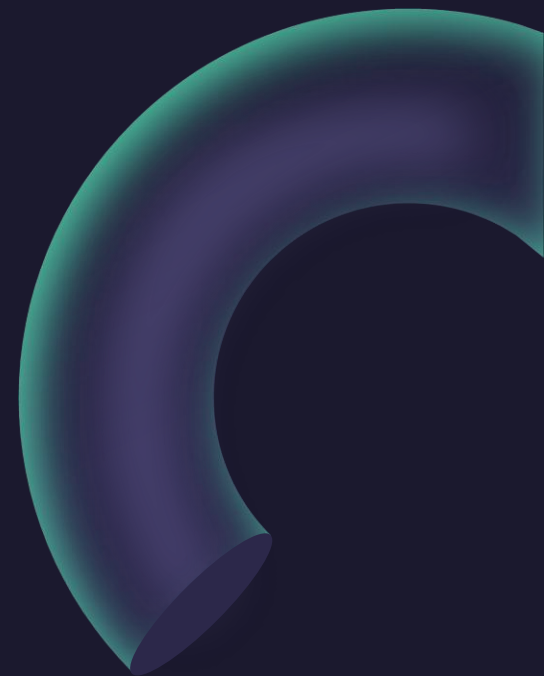
then

$$\iint_D f(x, y) \, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) \, dx \, dy$$

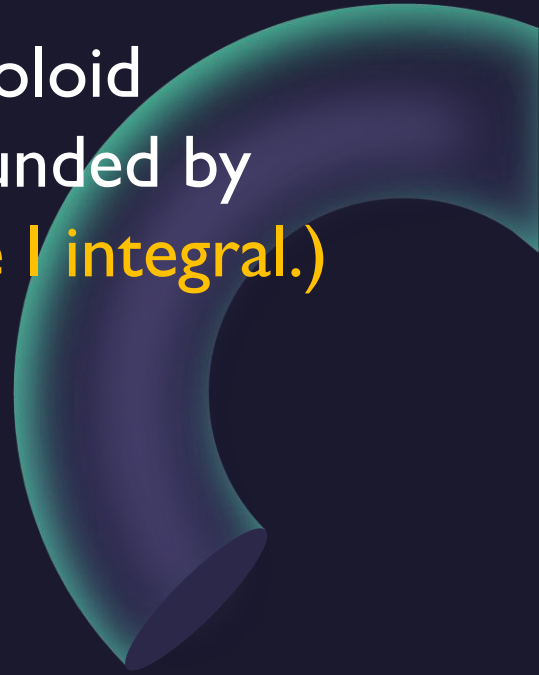
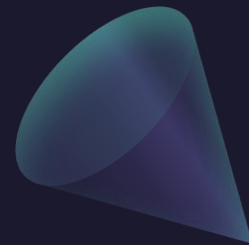
**Example:** Evaluate the double integral

$$\iint_R (x + 2y) \, dA$$

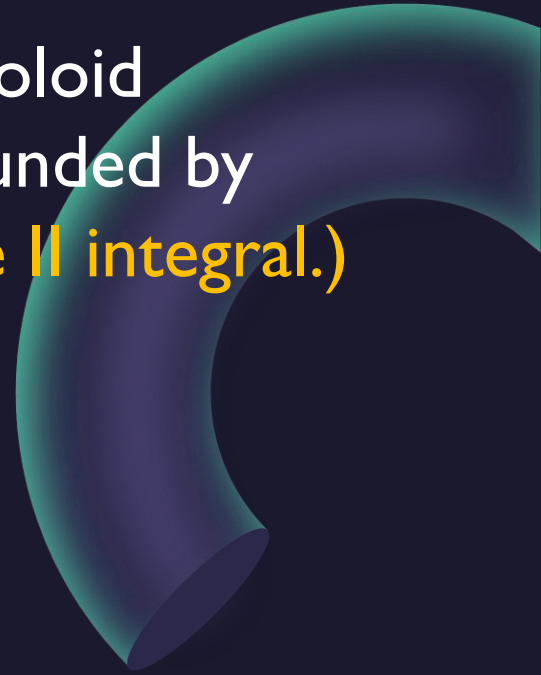
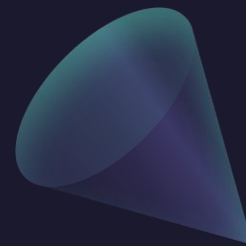
where  $R$  is the region bounded by the parabolas  
 $y = 2x^2$  and  $y = 1 + x^2$ .



**Example:** Find the volume of the solid that lies under the paraboloid  $z = x^2 + y^2$  and above the region in the  $xy$ -plane bounded by the line  $y = 2x$  and the parabola  $y = x^2$ . (As a Type I integral.)



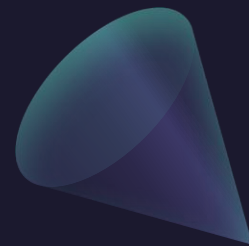
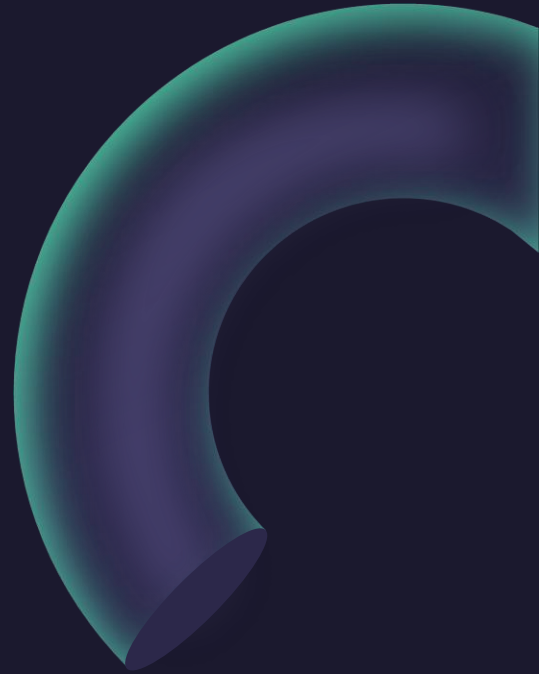
**Example:** Find the volume of the solid that lies under the paraboloid  $z = x^2 + y^2$  and above the region in the  $xy$ -plane bounded by the line  $y = 2x$  and the parabola  $y = x^2$ . (As a Type II integral.)



**Example:** Evaluate the iterated integral

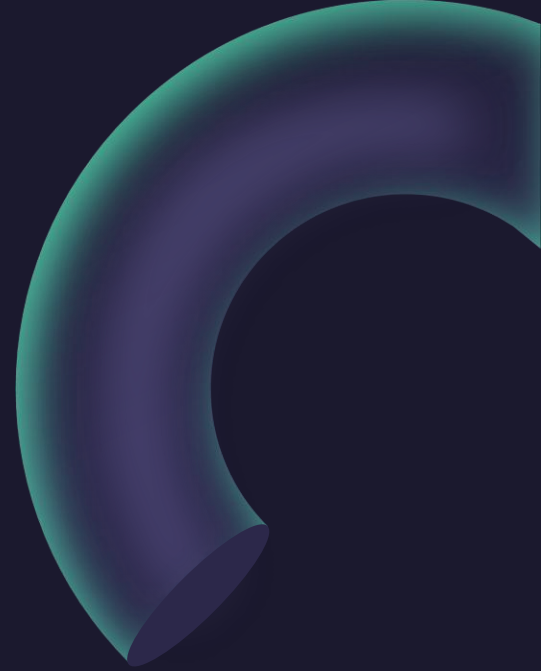
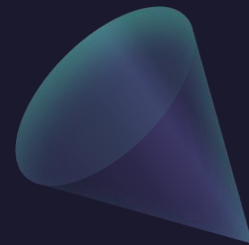
$$\int_0^1 \int_x^1 \sin(y^2) dy dx$$

$$\underbrace{\int \sin(y^2)}$$



**Example:** Evaluate the iterated integral

$$\int_0^1 \int_x^1 \sin(y^2) \, dy \, dx$$



# Properties of Double Integrals

$$\iint_D [f(x, y) + g(x, y)] dA = \iint_D f(x, y) dA + \iint_D g(x, y) dA$$

$$\iint_D c f(x, y) dA = c \iint_D f(x, y) dA \quad \text{where } c \text{ is a constant}$$

If  $f(x, y) \geq g(x, y)$  for all  $(x, y)$  in  $D$ , then

**7**

$$\iint_D f(x, y) dA \geq \iint_D g(x, y) dA$$

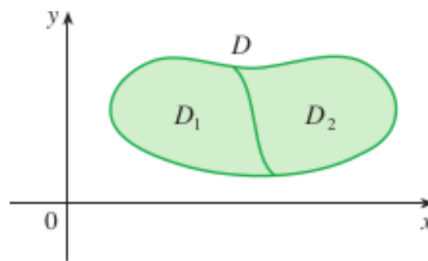
# Properties of Double Integrals

If  $D = D_1 \cup D_2$ , where  $D_1$  and  $D_2$  don't overlap except perhaps on their boundaries (see [Figure 17](#)), then

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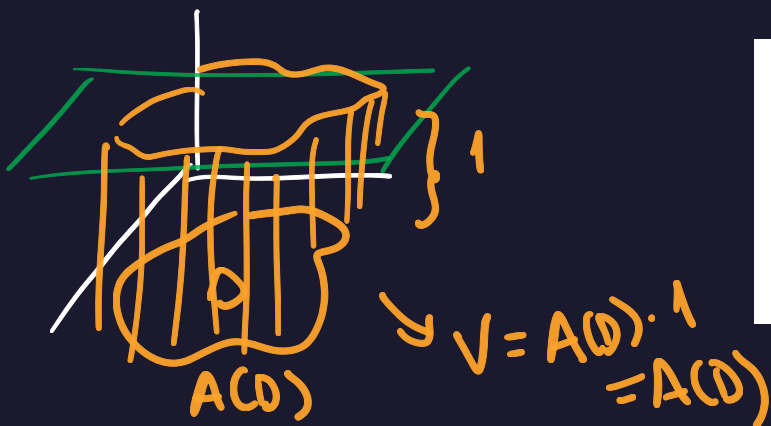
$$\iint_D f(x, y) \, dA = \iint_{D_1} f(x, y) \, dA + \iint_{D_2} f(x, y) \, dA$$

**Figure 17**

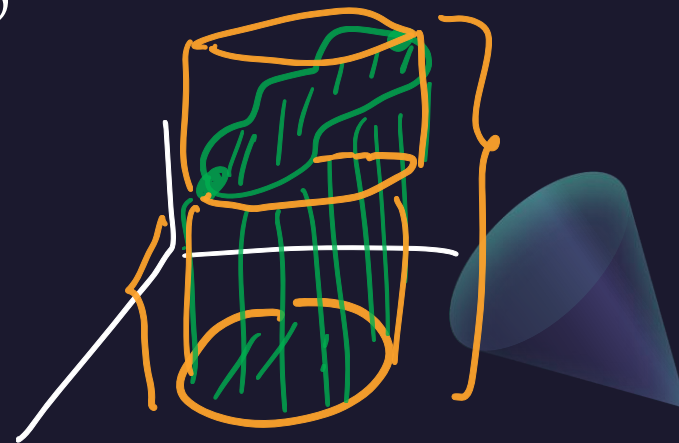




# Properties of Double Integrals



$$\iint_D 1 \, dA = A(D)$$



**10** If  $m \leq f(x, y) \leq M$  for all  $(x, y)$  in  $D$ , then

$$m \cdot A(D) \leq \iint_D f(x, y) \, dA \leq M \cdot A(D)$$

**Example:** Estimate the value of the double integral

$$\iint_R e^{-(x^2+y^2)} dA$$

where  $R = \{(x, y): x^2 + y^2 \leq 1\}$  is the circle of radius 1.

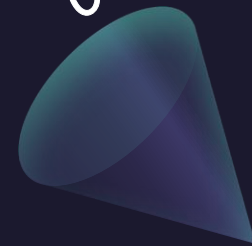
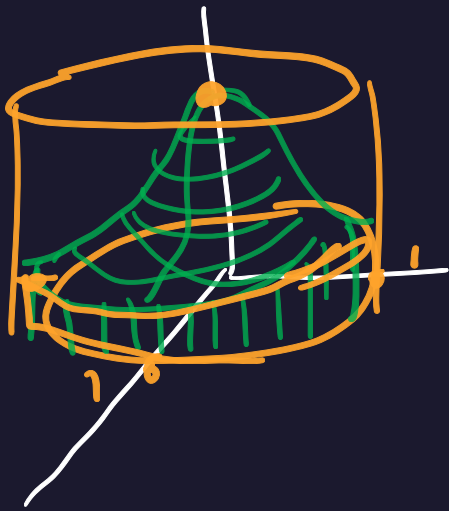
• MAX  $\nearrow f(x, y) = e^{-(x^2+y^2)}$

• MIN  $\nearrow$

in region  $R$ , circle of radius  $\leq 1$

methods  $\leadsto (0, 0)$  local max  $f(0, 0) = e^{-(0^2+0^2)} = e^{-0}$

MIN.  $x^2 + y^2 = 1$



$$= 1$$

# Questions?

$$f(x,y) = e^{-(x^2+y^2)} \quad \begin{matrix} \swarrow x^2+y^2=1 \\ = e^{-1} = 1/e \end{matrix} \quad \begin{matrix} \swarrow \text{Min} \\ \text{in } R \end{matrix}$$

$$\frac{1}{e} \cdot \pi \leq \iint_R e^{-(x^2+y^2)} dA \leq 1 \cdot \pi = \pi$$

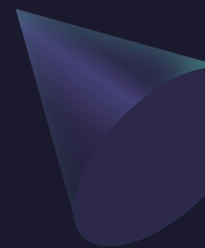
$$\frac{\pi}{e} \leq \iint_R e^{-(x^2+y^2)} dA \leq \pi$$

$$R = \{x^2+y^2 \leq 1\}$$

$$A(R) = \pi \cdot 1^2 = \pi$$



*ALVARO:* Start the recording!



# “Calculus 3”

## Multi-Variable Calculus

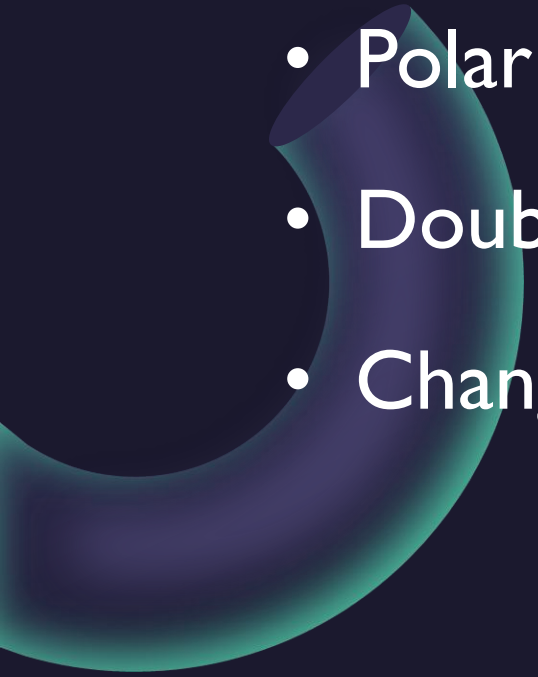
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## Double Integrals in Polar

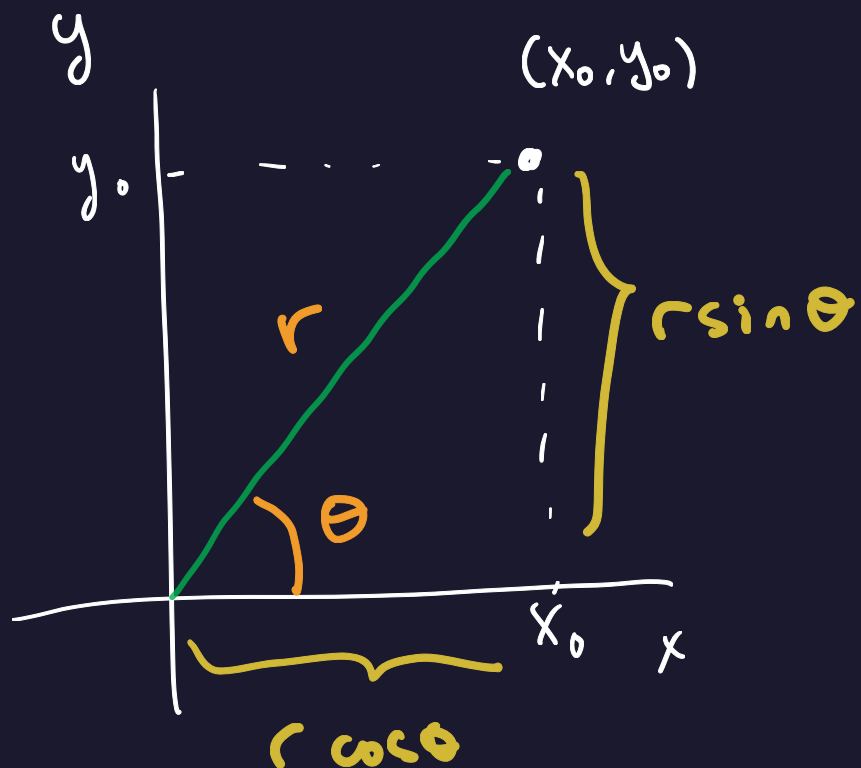




# Today – Double Integrals in Polar Coordinates!

- 
- Polar Coordinates
  - Double Integrals in Polar Coordinates
  - Change from Cartesian to Polar

# Polar Coordinates vs Cartesian



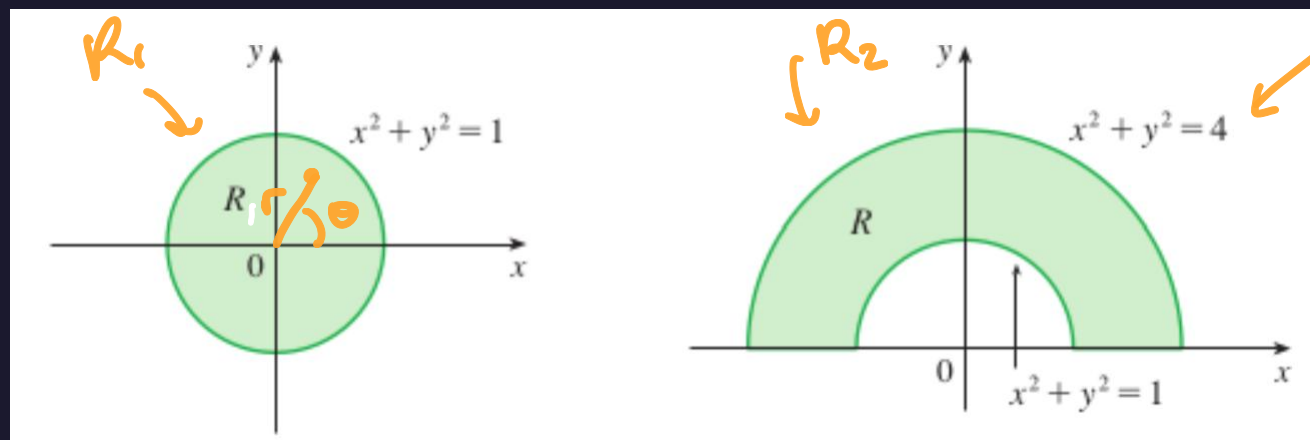
$$\begin{cases} x_0 = r \cos \theta \\ y_0 = r \sin \theta \end{cases}$$

$$r = \sqrt{x_0^2 + y_0^2}$$

$$r^2 = x_0^2 + y_0^2$$

$$(x_0, y_0) \longleftrightarrow (r, \theta)$$

**Example:** Express the following regions in polar coordinates.

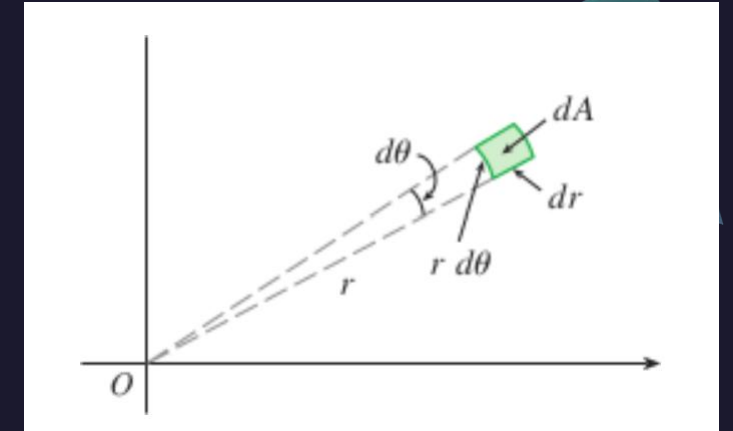
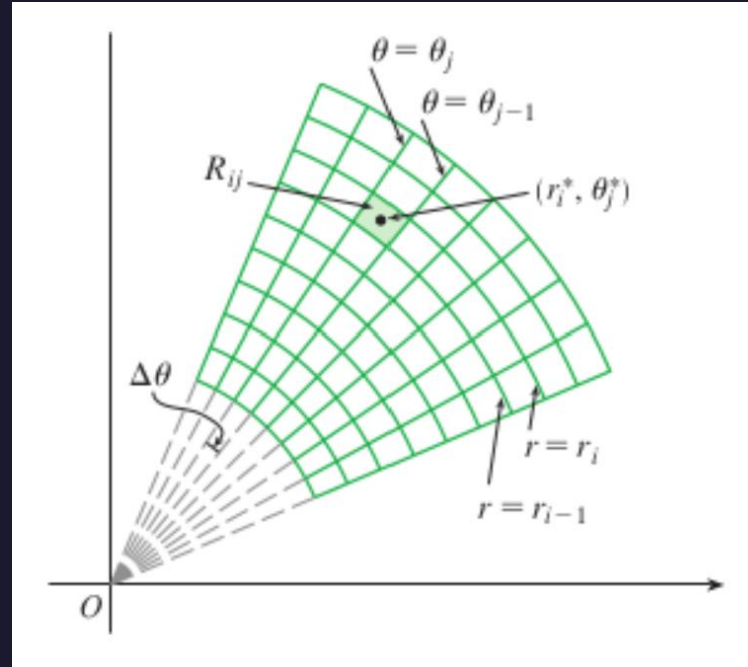
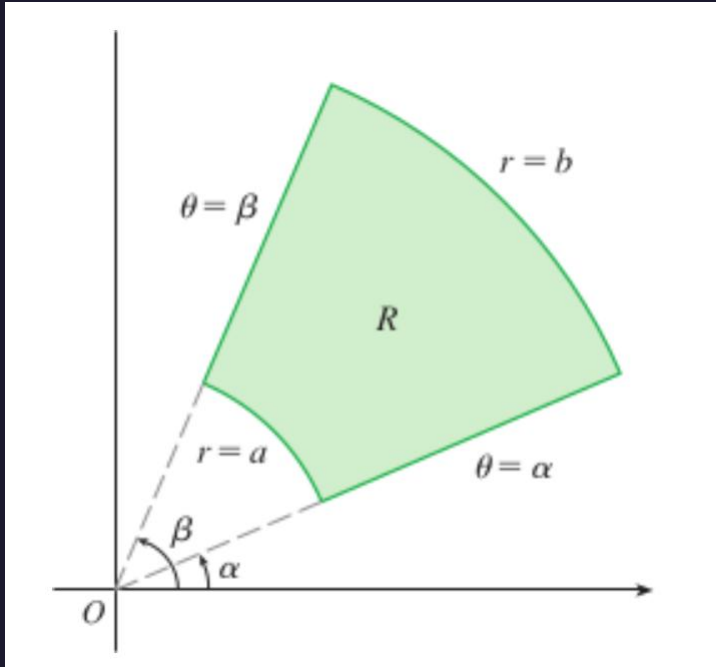


$$R_1 = \{(x, y) : x^2 + y^2 \leq 1\} = \{(r, \theta) : 0 \leq \theta \leq 2\pi, 0 \leq r \leq 1\}$$

$$R_2 = \{(x, y) : 1 \leq \overbrace{x^2 + y^2}^{r^2} \leq 4, y \geq 0\} = \{(r, \theta) : 0 \leq \theta \leq \pi, 1 \leq r \leq 2\}$$



# Double Integrals in Polar Coordinates

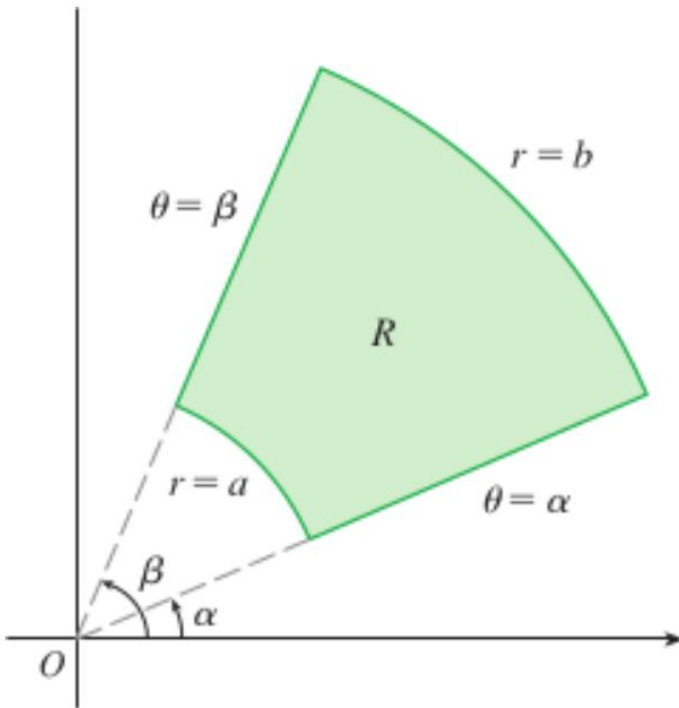


$$\sum_{i=1}^m \sum_{j=1}^n f(r_i^* \cos \theta_j^*, r_i^* \sin \theta_j^*) \Delta A_i = \sum_{i=1}^m \sum_{j=1}^n f(r_i^* \cos \theta_j^*, r_i^* \sin \theta_j^*) r_i^* \Delta r \Delta \theta$$

# Double Integrals in Polar Coordinates

If  $f$  is continuous on a polar rectangle  $R$  given by  $0 \leq a \leq r \leq b$ ,  $\alpha \leq \theta \leq \beta$ , where  $0 \leq \beta - \alpha \leq 2\pi$ , then

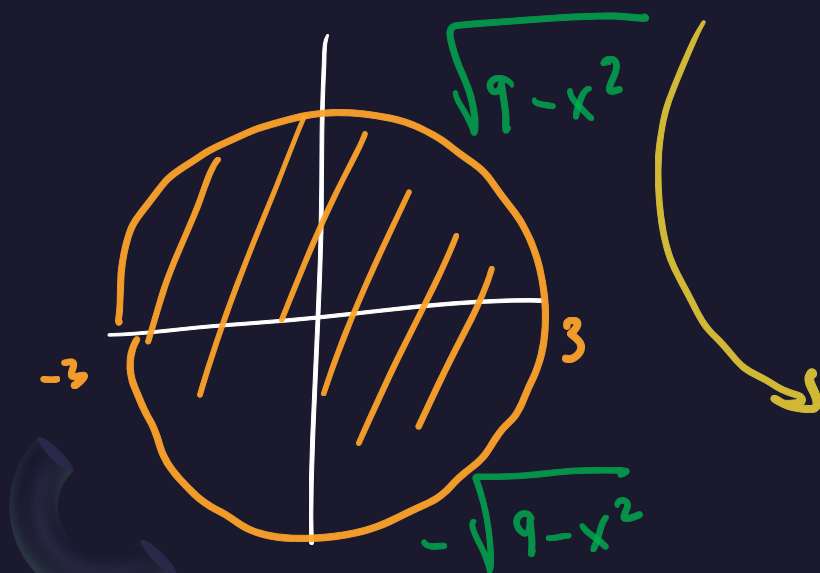
$$\iint_R f(x, y) dA = \int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) r dr d\theta$$



**Example:** Find the area of a circle of radius 3 using double integrals.

$$R = \{(x, y) : x^2 + y^2 \leq 9\}, \quad A(R) = \pi \cdot 3^2 = 9\pi$$

$$A(R) = \boxed{\iint_R 1 \cdot dA} = \int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} 1 \, dy \, dx$$



$$R = \{(r, \theta) : 0 \leq r \leq 3, 0 \leq \theta \leq 2\pi\}$$

$$f(x, y) = 1$$

$$\boxed{\int_0^{2\pi} \int_0^3 1 \cdot r \cdot dr \, d\theta}$$

**Example:** Find the area of a circle of radius 3 using double integrals.

$$\int_0^{2\pi} \int_0^3 r \, dr \, d\theta = \int_0^{2\pi} \left. \frac{r^2}{2} \right|_0^3 d\theta$$

$$= \int_0^{2\pi} \left( \frac{9}{2} - \frac{0}{2} \right) d\theta = \int_0^{2\pi} \frac{9}{2} d\theta$$

$$= \frac{9}{2} \theta \Big|_0^{2\pi} = \cancel{\frac{9}{2}} \cdot \cancel{2}\pi - \cancel{\frac{9}{2}} 0 = \boxed{9\pi}$$

$$\pi \cdot 3^2$$



**Example:** Find the area of a circle of radius  $R$  using double integrals.

$$\iint_R 1 \, dA \quad R = \{ (x, y) : x^2 + y^2 \leq R^2 \}$$
$$= \{ (r, \theta) : 0 \leq \theta \leq 2\pi, 0 \leq r \leq R \}$$

$$\left( \int_0^{2\pi} \int_0^R 1 \cdot r \, dr \, d\theta = \int_0^{2\pi} \left. \frac{r^2}{2} \right|_0^R d\theta \right.$$

$$= \int_0^{2\pi} \frac{R^2}{2} d\theta = \left. \frac{R^2}{2} \theta \right|_0^{2\pi} = \cancel{\frac{R^2}{2}} \cdot 2\pi - \cancel{\frac{R^2}{2}} \cdot 0$$

$$= \boxed{\pi R^2}$$

**Example:** Evaluate the value of the double integral

$$\iint_R (3x + 4y^2) dA$$

where  $R$  is the region in the upper half plane bounded by the circles of radius 1 and 2.

$$R = \{ (r, \theta) : 1 \leq r \leq 2, 0 \leq \theta \leq \pi \}$$

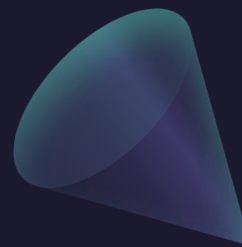
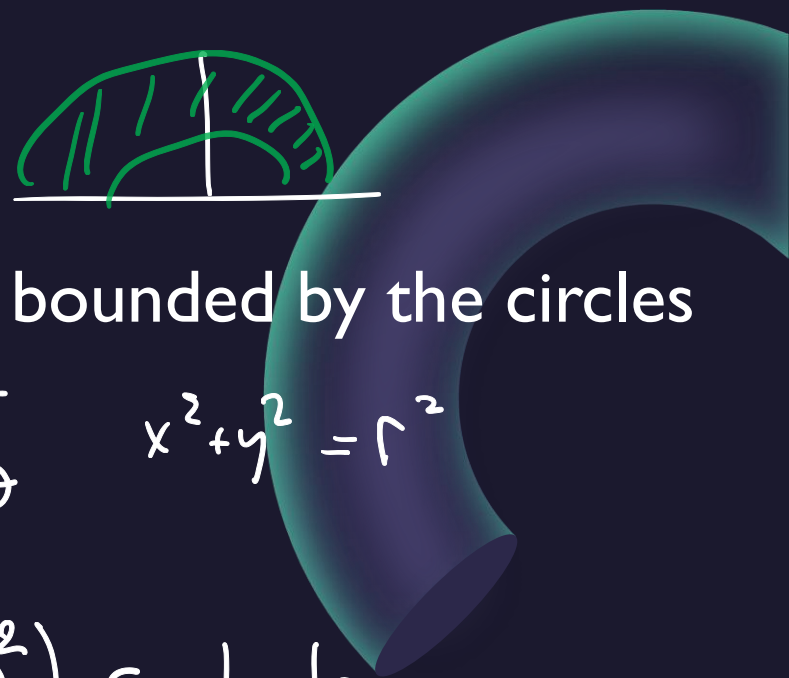
$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

$$x^2 + y^2 = r^2$$

$$\iint_R (3x + 4y^2) dA = \int_0^\pi \int_1^2 (3(r \cos \theta) + 4(r \sin \theta)^2) \cdot r \cdot dr d\theta$$

$$= \int_0^\pi \int_1^2 (3r^2 \cos \theta + 4r^3 \sin^2 \theta) dr d\theta = \int_0^\pi \left[ r^3 \cos \theta + r^4 \sin^2 \theta \right]_1^2 d\theta$$

$$= \int_0^\pi 7 \cos \theta + 15 \sin^2 \theta d\theta$$



**Example:** Evaluate the value of the double integral

$$\iint_R (3x + 4y^2) dA$$

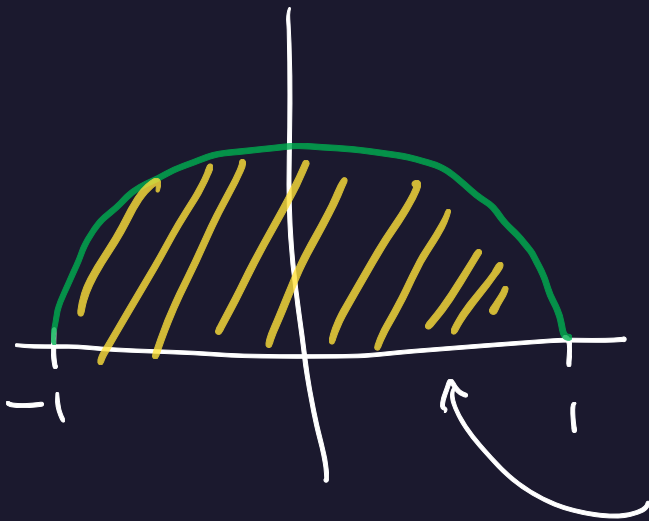
where  $R$  is the region in the upper half plane bounded by the circles of radius 1 and 2.

$$\begin{aligned}
 &= \int_0^{\pi} 7 \cos \theta + 15 \sin^2 \theta \, d\theta = \int_0^{\pi} 7 \cos \theta + \frac{15}{2} (1 - \cos 2\theta) \, d\theta \\
 &= 7 \sin \theta + \frac{15}{2} \theta - \frac{15}{4} \sin 2\theta \Big|_0^{\pi} = \boxed{\frac{15\pi}{2}}
 \end{aligned}$$

$$\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2}$$

**Example:** Evaluate the value of the double integral

$$I = \int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2) dy dx$$



$$y = \sqrt{1-x^2} \rightarrow y^2 = 1-x^2$$
$$y = 0 \quad \text{or} \quad x^2 + y^2 = 1$$

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ r^2 = x^2 + y^2 \end{cases}$$

$$R = \{(r, \theta) : 0 \leq \theta \leq \pi, 0 \leq r \leq 1\}$$

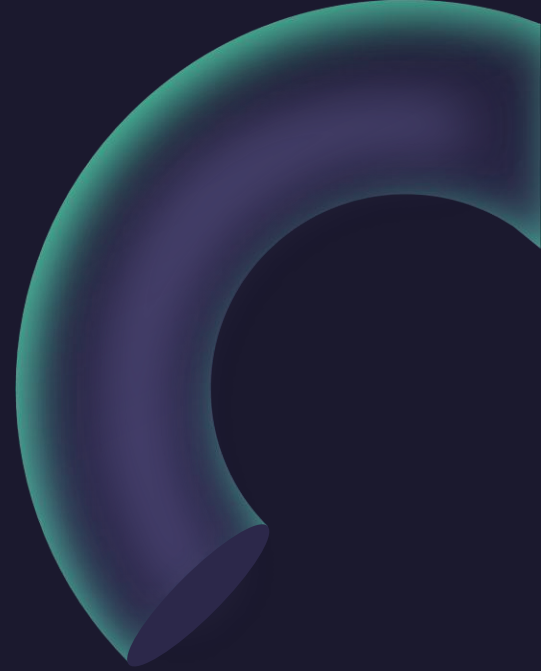
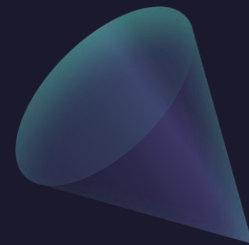
$$I = \int_0^\pi \int_0^1 r^2 \cdot r dr d\theta = \int_0^\pi \int_0^1 r^3 dr d\theta$$

$$= \int_0^\pi \left[ \frac{r^4}{4} \right]_0^1 d\theta = \int_0^\pi \frac{1}{4} d\theta = \frac{1}{4} \theta \Big|_0^\pi = \boxed{\frac{\pi}{4}}$$



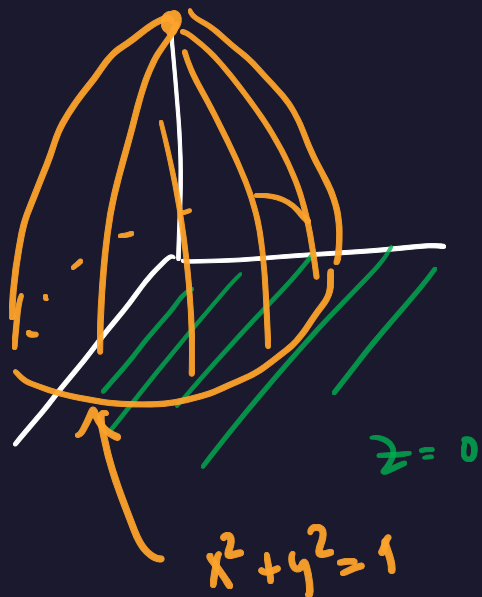
**Example:** Evaluate the value of the double integral

$$\int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2) dy dx$$



**Example:** Find the volume of the solid bounded by  $z = 0$  and the paraboloid

$$\left\{ \begin{array}{l} z = 1 - x^2 - y^2 \\ f(x, y) \\ z = 0 \end{array} \right\} \quad \left\{ \begin{array}{l} 0 = 1 - x^2 - y^2 \\ \Rightarrow x^2 + y^2 = 1 \end{array} \right.$$



$$R = \{(x, y) : x^2 + y^2 \leq 1\}$$

$$= \{(r, \theta) : 0 \leq \theta \leq 2\pi, 0 \leq r \leq 1\}$$

$$\iint_R 1 - x^2 - y^2 \, dA = \int_0^{2\pi} \int_0^1 (1 - r^2) \cdot r \, dr \, d\theta$$

$$\begin{aligned} &\uparrow \\ &1 - (x^2 + y^2) \\ &= 1 - r^2 \end{aligned}$$

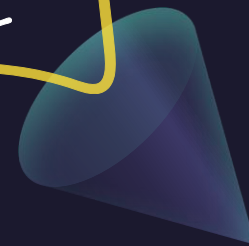
**Example:** Find the volume of the solid bounded by  $z = 0$  and the paraboloid  $z = 1 - x^2 - y^2$ .

$$\int_0^{2\pi} \int_0^1 (1 - r^2) \cdot r \, dr \, d\theta = \int_0^{2\pi} \int_0^1 r - r^3 \, dr \, d\theta$$

$$= \int_0^{2\pi} \left. \frac{r^2}{2} - \frac{r^4}{4} \right|_0^1 d\theta = \int_0^{2\pi} \frac{1}{2} - \frac{1}{4} \, d\theta$$

$$= \int_0^{2\pi} \frac{1}{4} \, d\theta = \left. \frac{\theta}{4} \right|_0^{2\pi} = \frac{2\pi}{4} - \frac{0}{4} = \boxed{\frac{\pi}{2}}$$

VOLUME

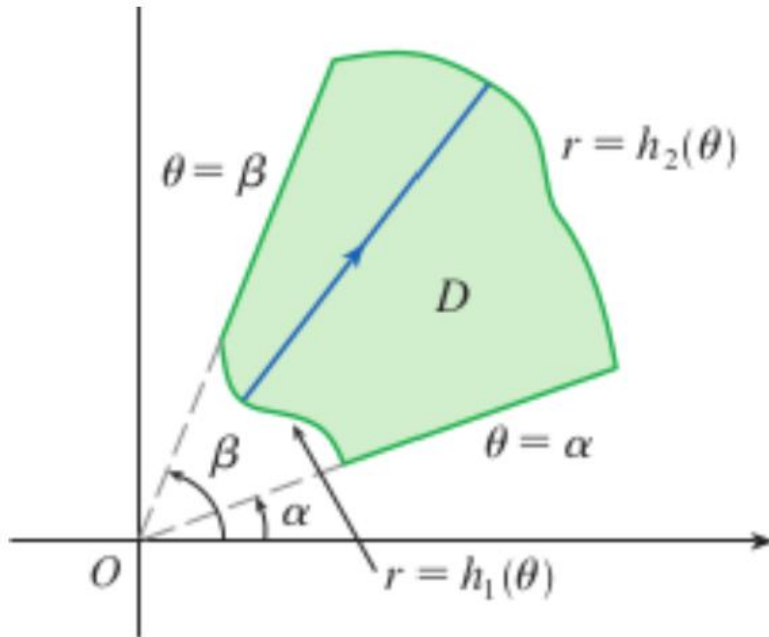


# Integrals in Polar Coordinates of Type I

$$D = \{(r, \theta) \mid \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\}$$

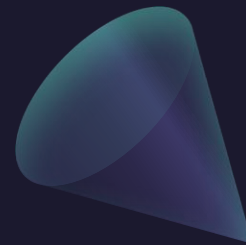
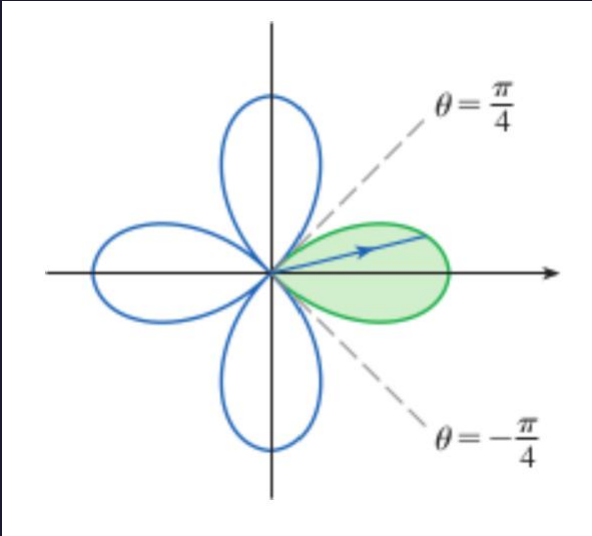
**3** If  $f$  is continuous on a polar region of the form

$$D = \{(r, \theta) \mid \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\}$$

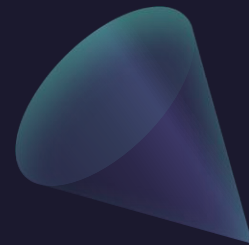


$$\iint_D f(x, y) \, dA = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} f(r \cos \theta, r \sin \theta) \, r \, dr \, d\theta$$

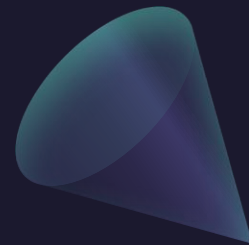
**Example:** Find the area enclosed by one loop of the four-leaved rose  
 $r = \cos(2\theta)$



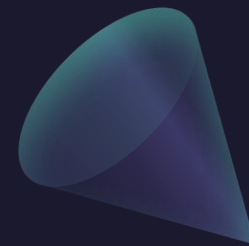
**Example:** Find the area enclosed by one loop of the four-leaved rose  
 $r = \cos(2\theta)$



**Example:** Find the volume of the solid that lies under the paraboloid  $z = x^2 + y^2$  above the  $xy$ -plane and inside the cylinder  $x^2 + y^2 = 2x$ .

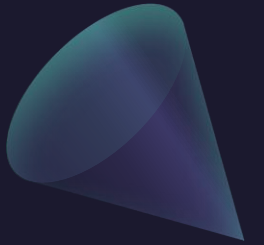


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# Questions?



# Thank you

Until next time.

